

AWS EC2 Disk Troubleshooting & EBS Storage Management

Today's DevOps Lab

1. Lab Objective

Understand how to troubleshoot a disk-full problem on an Ubuntu EC2 instance and increase or add storage using Amazon EBS.

Topics: disk usage, exact large files, inode usage, deleted-but-open files, Docker storage, EBS, mounting, increasing an existing EBS volume, adding a new EBS volume, formatting, mounting, and verification.

2. EC2 Storage Architecture

A typical EC2 instance uses an EBS volume as persistent block storage.

```
AWS
|
EC2 Instance
|
EBS Volume
|
Linux Filesystem
|
/
```

3. Check Disk Usage

The first command to run when the server reports a disk-full problem is:

```
df -h
```

Example:

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/root	20G	20G	0	100%	/

This means the root filesystem is completely full.

4. Check Filesystem Type

```
df -hT
```

Example:

Filesystem	Type	Size	Used	Avail	Use%	Mounted on
/dev/root	ext4	20G	20G	0	100%	/

The filesystem type matters when expanding the filesystem.

5. Find Which Directory Uses Space

```
sudo du -sh /* 2>/dev/null
```

Example:

```
500M    /home
```

```
12G    /var
4G     /usr
1G     /opt
200M   /tmp
```

If /var is largest, investigate /var next.

6. Drill Down

```
sudo du -sh /var/* 2>/dev/null
```

Example:

```
10G    /var/log
1.5G    /var/lib
300M    /var/cache
100M    /var/tmp
```

Then investigate /var/log.

7. Find the Exact File

```
sudo du -sh /var/log/* 2>/dev/null
```

Example:

```
6.5G    /var/log/myapp.log
2.1G    /var/log/syslog
900M    /var/log/auth.log
```

Now the exact large file is /var/log/myapp.log.

8. Find Large Files Directly

```
sudo find / -xdev -type f -size +500M -exec ls -lh {} \; 2>/dev/null
```

This searches the root filesystem for regular files larger than 500 MB.

9. Find the Largest Items

```
sudo du -ahx / | sort -rh | head -20
```

This displays the largest files and directories first.

10. Check Inode Usage

```
df -i
```

If IUse% is 100%, the filesystem may have run out of inodes because of too many small files.

11. Deleted Files Still Using Disk

```
sudo lsof +L1
```

A deleted file can continue consuming disk space while a running process still has it open.

```
ps -fp <PID>
```

Identify the process and restart the responsible service appropriately.

12. Check Systemd Journal Logs

```
journalctl --disk-usage
sudo journalctl --vacuum-time=7d
sudo journalctl --vacuum-size=500M
```

Use cleanup commands carefully and according to your log-retention requirements.

13. Check Ubuntu Package Cache

```
sudo du -sh /var/cache/apt
sudo apt clean
sudo apt autoremove
```

Verify the result with `df -h`.

14. Check Docker Storage

```
docker system df
docker system df -v
sudo du -sh /var/lib/docker
docker ps -a
docker images
docker volume ls
```

Unused resources can be cleaned carefully with `docker system prune`. Do not blindly use aggressive cleanup on production systems.

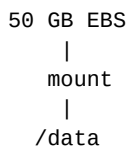
15. What Is EBS?

EBS means Elastic Block Store. It provides persistent block storage for EC2 instances.



16. What Is Mounting?

Mounting means making a filesystem available through a directory in Linux.



After mounting, files created under `/data` are stored on that filesystem.

17. Attach vs Mount

```
ATTACH = EBS Volume -> EC2
MOUNT = Disk/Filesystem -> Linux directory
```

Attach is an AWS operation. Mount is a Linux filesystem operation.

18. Increase an Existing EBS Volume

Example: increase the root EBS volume from 20 GB to 40 GB.

```
df -h
lsblk
```

In AWS Console: EC2 -> Elastic Block Store -> Volumes -> select volume -> Actions -> Modify volume -> change size -> Modify.

After AWS increases the volume, Linux may still show the old partition/filesystem size.

19. Extend the Partition

```
sudo apt update
sudo apt install -y cloud-guest-utils
sudo growpart /dev/nvme0n1 1
lsblk
```

Always verify the actual device and partition with lsblk before running growpart.

20. Extend the Filesystem

For Ext4:

```
sudo resize2fs /dev/nvme0n1p1
```

For XFS, use the appropriate xfs_growfs command for the mounted filesystem. Check the filesystem type with df -hT first.

```
df -h
```

21. Add a New EBS Volume

Create a new EBS volume in the same Availability Zone as the EC2 instance, then attach it to the instance.

```
EC2
|
+-- Root EBS 20 GB -> /
|
+-- New EBS 50 GB -> /data
```

22. Find the New Disk

```
lsblk
```

Example:

NAME	SIZE	TYPE	MOUNTPPOINT
nvme0n1	20G	disk	
■ nvme0n1p1	20G	part	/
nvme1n1	50G	disk	

The new disk may be /dev/nvme1n1 on a Nitro-based instance. Always confirm with lsblk.

23. Format and Mount a New Empty Disk

```
sudo mkfs.ext4 /dev/nvme1n1
sudo mkdir /data
sudo mount /dev/nvme1n1 /data
df -h
```

WARNING: mkfs can destroy existing data on the selected device. Only format a confirmed new/empty volume.

24. Persistent Mount

```
sudo blkid /dev/nvme1n1
sudo mount -a
df -h
```

Use the filesystem UUID in /etc/fstab for persistent mounting. Test the fstab configuration with mount -a before rebooting.

25. Complete Real-World Disk-Full Flow

```
EC2 Disk Full
|
v
df -h
|
v
sudo du -sh /*
|
v
Drill down with du
|
v
Find exact file
|
v
ls -lh /path/to/file
|
v
sudo lsof /path/to/file
|
v
Fix root cause
|
+--> Log rotation / safe cleanup / Docker cleanup
|
+--> Increase EBS if more capacity is required
|
v
df -h
|
v
VERIFY
```

26. Important Commands

```
df -h
df -hT
lsblk
sudo du -sh /*
sudo du -sh /var/*
sudo find / -xdev -type f -size +500M -exec ls -lh {} \; 2>/dev/null
sudo du -ahx / | sort -rh | head -20
df -i
sudo lsof +L1
journalctl --disk-usage
docker system df
sudo growpart /dev/nvme0n1 1
sudo resize2fs /dev/nvme0n1p1
sudo blkid
sudo mount /dev/nvme1n1 /data
sudo umount /data
```

27. Interview Answer

If a production EC2 server is 100% full, first run `df -h` to identify the full filesystem. Then use `du` to find the directory consuming the most space and drill down to the exact large file. Use `lsof` to check whether a process is still using the file and `df -i` to check inode exhaustion. If Docker is installed, check Docker storage. Safely clean or rotate unnecessary data, identify the root cause, and verify with `df -h`. If more capacity is required, increase the EBS volume or attach a new EBS volume, then extend the partition and filesystem.

28. Key Takeaways

DISK FULL

```
|  
+--> df -h  
+--> du  
+--> find exact file  
+--> lsof  
+--> fix root cause
```

ADD STORAGE

```
|  
+--> EBS  
+--> Attach to EC2  
+--> Format new disk if required  
+--> Mount  
+--> Verify with df -h
```

INCREASE EXISTING EBS

```
|  
+--> Modify EBS size  
+--> growpart  
+--> resize filesystem  
+--> df -h
```